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# From Sources to Networks. The TRACES Project and the Reconstruction of Mobility Systems from Antiquity to the Middle Ages

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## Abstract

This paper presents the TRACES project, funded within the GALILEO2026 programme and developed through an international, interdisciplinary collaboration between the University of Cassino and Southern Lazio – European University of Technology and the Université de Versailles Saint-Quentin-en-Yvelines. The main argument is that network thinking fundamentally reshapes the interpretation of historical and archaeological sources, moving beyond data extraction to enable a deeper understanding of past mobility systems. By framing mobility, circulation, and connectivity as networked phenomena, TRACES explicitly engages with both the opportunities and limitations posed by heterogeneous and diachronically stratified sources for historical network research.

The project investigates communication systems from Roman Gaul to the post-medieval period across a wide geographical area largely corresponding to present-day France. Its methodological core is the integration of diverse categories of sources (archaeological remains, cartographic documents, ancient itineraries, metal objects and textual evidence) within a shared GIS and network-analytical framework. These sources are modelled into complementary network layers (planned, infrastructural, material, and informational), which are then integrated into multiplex network models to capture interactions among routes, objects, and human mobility.

Particular attention is devoted to networks reconstructed from ancient itineraries (e.g., the *Tabula Peutingeriana* and the *Itinerarium Antonini*), medieval mortuary rolls, early modern cartography, the spatial distribution of artefacts, and textual sources.

The project also integrates structured archaeological datasets and interoperable digital repositories, including geospatial and analytical databases designed according to FAIR and Linked Open Data principles. Particular attention is devoted to the reuse of relational and spatially enabled archaeological datasets, such as the CHIPS database for archaeometallurgical evidence, whose PostgreSQL/PostGIS architecture and standardised geospatial descriptors provide a useful methodological reference for the harmonisation, interoperability, and comparative analysis of heterogeneous historical sources.

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The contribution discusses challenges in translating these sources into network data and reflects on how abstraction influences interpretation.

The paper also presents the first phase of the project's implementation, including the design of a WebGIS platform and the initial GIS-based analyses (cost-surface modelling, least-cost paths, and network analysis). Preliminary results illustrate how postdictive approaches and the introduction of cultural and historical attractors can mitigate source-related biases and enhance the heuristic value of network models.

By combining methodological reflection with early empirical results, this contribution demonstrates that network thinking transforms historical source interpretation through integrated modelling and analysis. It argues that a source-aware approach is essential for advancing historical network research.

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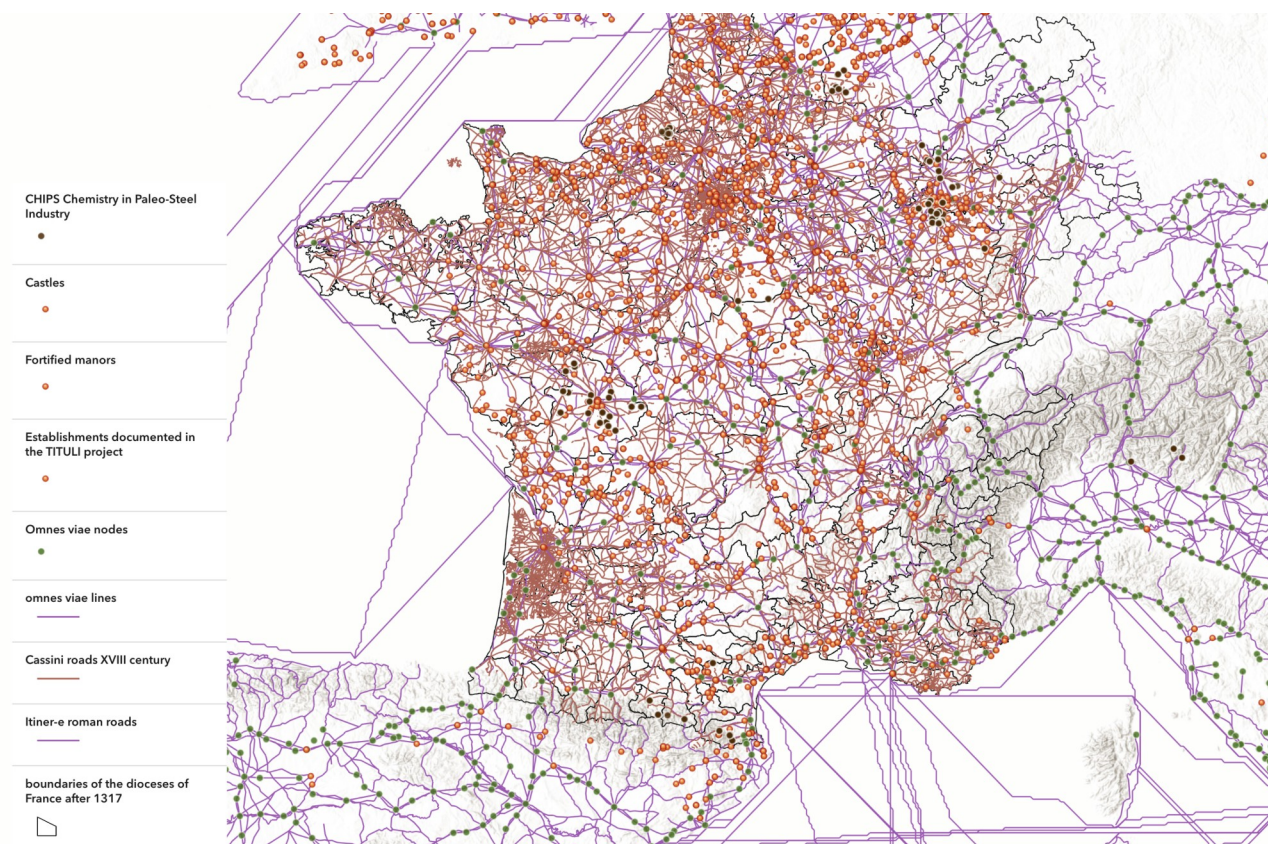


Figure 1. Preliminary integration of heterogeneous historical and archaeological datasets used within the TRACES framework. Source: authors' elaboration.